```
#1.. Explain the difference between correlation and causation in this s
# Correlation:

- Correlation means two variables are related to each other.
- In this situation, customers who receive more promotional emails tend
- It shows a positive relationship between emails and purchases.
Causation:
- Causation means one variable directly causes a change in another vari
- The company has not proved that sending more emails directly increase
- Other factors may also influence customer purchases.
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# 2..Identify two possible confounding variables.
Two possible confounding variables are:

1. Customer Interest: Customers who are more interested in products may rece

2. Previous Purchase History: Regular customers may receive more promotional
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# 3..Explain why the company's conclusion may be incorrect.
The company assumes that more promotional emails always increase sales.
However, correlation does not prove causation.
Other factors, such as customer interest and previous purchase history, may
Sending too many emails may annoy customers.
Some customers may unsubscribe or ignore promotional emails.
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# 4.Suggest an experiment to estimate the causal effect of promotional email
Divide customers randomly into two groups.
Group A (Control Group): Send fewer or no promotional emails.
Group B (Treatment Group): Send more promotional emails.
Keep the experiment duration and other marketing conditions similar.
Measure the number of purchases made by both groups.
Compare the purchase rates of both groups.
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# 5..Explain how the company could use the results to make better marketing
Identify the number of promotional emails that generate better sales.
Avoid sending unnecessary emails to customers.
Focus on customers who are interested in the products.
Reduce email frequency for customers who unsubscribe or ignore emails.
Use experimental results to improve marketing strategies.
Increase sales while maintaining better customer satisfaction.
```

2 A real-estate company builds a Linear Regression model using the following features:

```
# 1.Identify the features that may cause multicollinearity.
The features Area_sqft and Area_sqm are the main cause of
multicollinearity because both represent the same house area in different un
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features, such as Bedrooms and Bathrooms, may also be correlated with house
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```
estimates.
ents bahut change ho sakte hain.
te hain.
na difficult ho jata hai.
ain.
in, lekin iska matlab ye nahi ki predictions hamesha inaccurate hongi.
```

```
# 3..Differentiate between perfect and high multicollinearity using this
Perfect Multicollinearity                          High Multicollinearity
Exact linear relationship hota hai.                Strong but non-exact re
Ek feature doosre se exactly calculate             Ek feature doosre se ex
ho sakta hai.
Correlation is +1 or -1.                          Correlation is close to
Regression coefficients cannot be uniquely        Coefficients can be est:
estimated using ordinary least squares.
Example: Area_sqft and Area_sqm.                    Example: Area_sqft and
```

```
# 4..Explain how VIF can be used to investigate the problem.
Variance Inflation Factor (VIF) is a statistical technique used to detec
variables in a regression model. It measures how much the variance of a
with other independent variables. The formula for VIF is:
FORMULA:->
            VIF = \frac{1}{1-R^2}

A VIF value of 1 indicates no multicollinearity, a value between 1 and 5
a value between 5 and 10 indicates high multicollinearity, and a value a

In the given real-estate dataset, VIF can be calculated for Area_sqft, A
 The VIF values of Area_sqft and Area_sqm are expected to be very high b
 different units. If a feature has a high VIF value, one of the highly c
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# 5..Explain how a correlation matrix can help detect multicollinearity
A correlation matrix is a statistical tool used to identify multicolline
variables in a regression model. It shows the strength and direction of
 The correlation coefficient ranges from -1 to +1. A value close to +1 i
  to -1 indicates a strong negative correlation. A value close to 0 indi
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# 6..Recommend appropriate ways to handle multicollinearity.
Multicollinearity can be handled using several techniques to improve the sta
 First, we can remove one of the highly correlated independent variables to
 real-estate dataset, we can remove either Area_sqft or Area_sqm because bot
 Second, we can use VIF (Variance Inflation Factor) to identify features wit
  (Principal Component Analysis) to combine correlated features into fewer u
  be applied to reduce the effect of multicollinearity by adding a penalty t
  we should calculate the correlation matrix and VIF again after removing or
```

3 Create a dataset containing at least four numerical features related to house prices.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from statsmodels.stats.outliers_influence import (
    variance_inflation_factor
)
```

```
# Create a Pandas DataFrame.
data = {
    "Area_sqft": [
        800, 1000, 1200, 1400, 1600],
    "Area_sqm": [
        74.32, 92.90, 111.48, 130.06, 148.64],
    # "Bedrooms": [
    #     2, 2, 3, 3, 3],
    # "Floors": [
    #     1, 1, 2, 2, 3]
}

df = pd.DataFrame(data)
```

```
# Calculate the correlation matrix.
correlation_matrics=df.corr()
x=df.copy()
```

```
# Calculate VIF for each predictor.
vif_data=pd.DataFrame()
vif_data['features']=x.columns
vif_data['VIF']=[
    variance_inflation_factor(x.values,i)for i in range(x.shape[1])
]
```

```
/tmp/ipykernel_2261/3590155737.py:5: UserWarning: The design matrix is poorly
  variance_inflation_factor(x.values,i)for i in range(x.shape[1])
```

```
# Interpret the VIF results.
print('\n VIF result:')
vif_data.round(2)
```

```
 VIF result:
```

	features	VIF
0	Area_sqft	1.000800e+15
1	Area_sqm	1.000800e+15

```
# Identify any redundant features.
redundant_features =["Area_sqm"]
```

```
df=df.drop(columns=redundant_features,errors='ignore')
print(df)
```

```
   Area_sqft
0        800
1       1000
2       1200
3       1400
4       1600
```

Start coding or generate with AI.

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